COMMON HALF YEARLY EXAMINATION - 2025

Standard XII

Reg.No. [ ][ ][ ][ ][ ][ ]

BIOLOGY

Time : 3.00 hrs | Part – I (Bio-Botany) 35 marks | Marks : 70

I. Choose the correct answer: 8 x 1 = 8

1. Male gametes is angiosperms are formed by the division of
   a) microspore mother cell b) microspore
   c) generative cell d) vegetative cell
2. Point mutation involves
   a) Deletion b) Insertion c) Change in single base pair d) duplication
3. In soil water available for plants is
   a) gravitational water b) chemically bound water
   c) capillary water d) hygroscopic water
4. For transformation, micro-particles coated with DNA to be bombarded with gene gun are made up of a) silver (or) platinum b) platinum (or) zinc
   c) silicon (or) platinum d) gold (or) tungsten
5. Find out the correctly matched pair.
   a) Rubber – Shorea robusta b) Dye – Indigofera annecta
   c) Timber – Cyperus papyrus d) Pulp – Hevea brasiliensis
6. Consumption of which one of the following foods prevent the kind of blindness associated with Vitamin A deficiency?
   a) Flavr savr b) Canola c) Golden rice d) Bt brinjal
7. Which of the following ecosystem has the highest primary productivity?
   a) Pond ecosystem b) Lake ecosystem
   c) Grassland ecosystem d) Forest ecosystem
8. With respect to Eichhornia
   Statement A : It drains off oxygen from water and is seen growing in standing water.
   Statement B : It is an indigenous species of our country
   a) Statement A is correct and statement B is wrong
   b) Both statements A and B are correct
   c) Statement A is wrong and statement B is correct
   d) Both statements A and B are wrong

II. Answer any four questions 4 x 2 = 8

9. What is Pollen kilt?
10. Write a note on heterosis?
11. Differentiate bio-medicines and botanical medicines?
12. Give four examples of plants cultivated in commercial agroforestry?
13. What is Atavism?
14. Draw the diagram of pollengrain and label the parts?

III. Answer any three questions. (Q.No.19 is compulsory) 3 x 3 = 9

15. Mention the name of man-made cereal. How it is formed?
16. What do you know about the word PBR 332?
17. Write the objectives of Plant Breeding.
18. Write the advantages of Artificial seeds.
19. What are Endemic Centres and Endemic Plants? Write any one example?

IV. Answer all the questions. 2 x 5 = 10

20. a) Enumerate the characteristic features of Entomophilous Flowers *(OR)*
    b) Explain the basic concepts involved in plant tissue culture.
21. a) What is Ten percent law? Explain with example? *(OR)*
    b) What are psychoactive drugs? Add a note Marijuana and Opium?

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TIMING : 1.30 Hrs COMMON HALF YEARLY EXAM-2025 (TOTAL MARK :35)
CLASS : XII PART – I – BIO-BOTANY
TENTATIVE ANSWER KEY

CHOOSE THE CORRECT ANSWER (8 × 1 = 8)

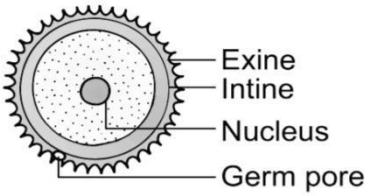
Q.NO	ANSWER	MARKS
1	(c) Generative cell	1
2	(c) Change in single base pair	1
3	(c) Capillary water	1
4	(d) Gold (or Tungsten	1
5	(b) Dye – *Indigofera annecta*	1
6	(c) Golden rice	1
7	(d) Forest ecosystem	1
8	(a) Statement A is correct and Statement B is wrong	1

PART – II **(4 × 2 = 8)**

Answer any FOUR from the following question.

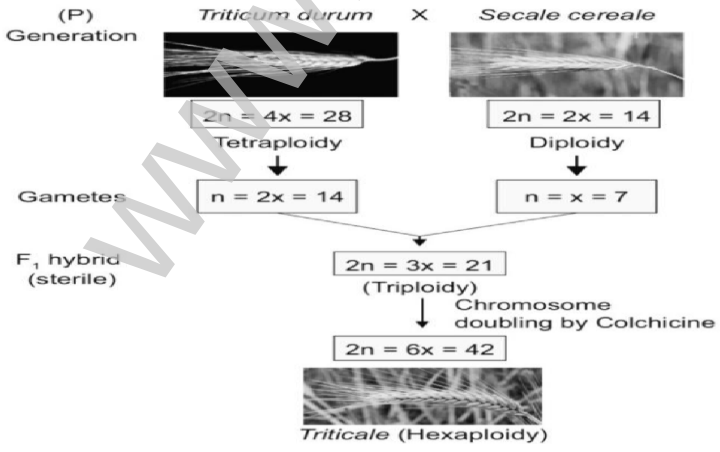
Q.NO	ANSWER	MARKS SPLITT	TOTAL MARKS
9	What is Pollen kit? i) It is an oily layer forming a thick viscous coating over pollen surface. ii) Pollenkitt is yellow or orange coloured and is chiefly made of carotenoids or flavonoids.	(2)	(2)
10	**Write a note on heterosis or Hybrid vigour** i) G.H. Shull first used the term heterosis. ii)The superiority of the F_1 hybrid in performance over its parents is called heterosis	(2)	(2)
11	**Differentiate bio-medicines and botanical medicines.** **Biomedicines** The medicinally useful molecules obtained from plants that are marketed as drugs are called Biomedicines. **Botanical medicines** Medicinal plants which are marketed as powders or in other modified forms are known as Botanical medicines.	(2)	(2)
12	**Give four examples of plants cultivated in commercial agroforestry.** ✦ Casuarina ✦ Eucalyptus ✦ Malai Vembu ✦ Teak	(2)	(2)

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	✦ Kadambu trees.		
13	What is Atavism ? i) Atavism is a modification of a biological structure. ii) An ancestral that reappears after having been lost through evolutionary changes in the previous generation	(2)	(2)
14	Write the advantages of Bt cotton?(Any 2) 	(1+1=2)	(2)

PART – III **(3 × 3 = 9)**

Answer any THREE from the following question. (Q.NO.19 is compulsory Question)

Q.NO	ANSWER	MARKS SPLITT	TOTAL MARKS
15	**Mention the name of man-made cereal. How it is formed?** i) Triticale is the successful first man made cereal. ii) Depending on the ploidy level Triticale can be divided into three main groups. **Tetraploidy** Crosses between diploid wheat and rye **Hexaploidy** Crosses between tetraploid wheat Triticum durum (macaroni wheat) and rye **Octoploidy** Crosses between hexaploid wheat T.aestivum (bread wheat) and rye Ex : They combine the high-protein content of wheat with rye‟s high content of the amino acid lysine, which is low in wheat. It can be explained by chart below 	(1) (1) (1)	(3)
16	**What do you know about the word pBR322?** • pBR 322 plasmid is a reconstructed plasmid. • It is most widely used as cloning vector.		

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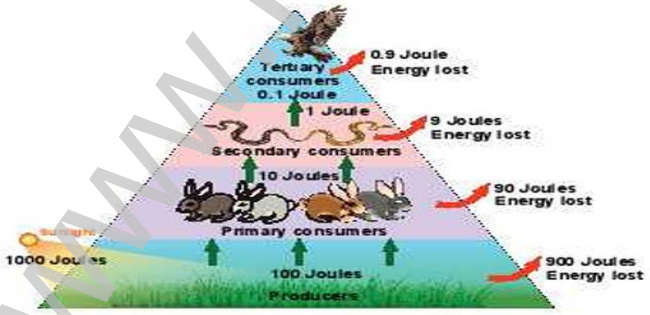
	• It contains 4361 base pairs. • In pBR, p for plasmid, B and R for scientist Boliver and Rodriguez who developed this plasmid. • The number 322 is the number of plasmid developed from their laboratory.	(1+1+1=3)	(3)
17	**Write the objectives of plant breeding** • To increase yield, vigour and fertility of the crop • To increase tolerance to environmental condition, salinity, temperature and drought. • To prevent the premature falling of buds, fruits etc.	(1) (1) (1)	(3)
18	**Write the advantages of artificial seeds.** i) Artificial seeds have many advantages over the true seeds. ii) Millions of artificial seeds can be produced at any time at low cost. iii) They provide an easy method to produce genetically engineered plants with desirable traits.	(1+1+1=3)	(3)
19 (C.Q)	**Endemic centres and Endemic plants** Endemic species are plants and animals that exist only in one geographic region. Species can be endemic to large or small areas of the earth. **Endemic plants** Some are endemic to a particular continent some to a particular continent and others to a single island. **E.g. Endemic center :** Southern and western Ghats **E.g.Endemic plants :** *Baccaurea courtallensis*	(1+1=2) (1)	(3)

PART – IV **(2 × 5 = 10)**

Answer ALL from the following question

Q.NO	ANSWER	MARKS SPLITT	TOTAL MARKS
20 (a)	**Enumerate the characteristic features of Entomophilous flowers.** **i)** Flowers are generally large. ii) If small they are aggregated in dense inflorescence. Ex: Asteraceae flowers. iii) Flowers are brightly coloured to attract insects. ex : Poinsettia and Bougainvillea iv) Flowers are scented and produce nectar. v)Pollen and nectar are the floral rewards for the visitors. vi) Flowers pollinated by flies and beetles produce foul odour to attract pollinators vii) In some flowers juicy cells are present which are pierced and the contents are sucked by the insects. xi) It contributes the cytoplasm during fertilization. Since the male gamete contribute only the nucleus but not cytoplasm.	(1+1+ 1+1 +1=5)	**(5)**

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20 (b)	**16. Explain the basic concepts involved in plant tissue culture.** ➢ **Basic concepts of plant tissue culture are** a) Totipotency b) Differentiation c) dedifferentiation d) redifferentiation **a) Totipotency** The property of live plant cells that they have the genetic potential when cultured in nutrient medium to give rise to a complete individual plant. **b) Differentiation** i) The process of biochemical and structural changes by which cells become specialized in form and function. ii) Meristematic tissue into mature cells. **c) Dedifferentiation** i) The reversion of mature cells to the meristematic state leading to the formation of callus is called dedifferentiation. ii) Mature cells into the meristematic tissue. iii) Formation of callus is called dedifferentiation. **d) Redifferentiation** The component cells of callus have the ability to form a whole plant in a nutrient medium is called redifferentiation	**(1)** **(1)** **(2)** **(1)**	**(5)**
21 (a)	**What is meant by Ten percent law?** • This law was proposed by Lindeman. • It states that during transfer of food energy from one trophic level to other, only about 10% is stored at every level and rest of them (90%) is lost in respiration, decomposition and in the form of heat. Hence, the law is called ten percent law. **Examples : Diagram or Explanation**  	**(1)** **(1)** **(1)** **(1)** **(1)**	**(5)**
21 (b)	**What are psychoactive drugs? Add a note Marijuana and Opium.** The phytochemicals / drugs from some of the plants alter an individual‟s perceptions of mind by producing hallucination are known as psychoactive drugs. **Opium** • Botanical name : Papaver somniferum • It is native to South Eastern Europe and Western Asia.	**(1)** **(1)**	**(5)**

	• Madhya Pradesh, Rajasthan and Uttar Pradesh are the licenced states to cultivate opium poppy. • It is derived from the exudates of fruits of poppy plants. • Opium was traditionally used to induce sleep and for relieving pain. • It yields Morphine, a strong analgesic which is used in surgery. • However, opium is an addiction forming drug. **Marijuana** • Botanical name **:** Cannabis sativa **Origin :** China • Gujarat, Himachal Pradesh, Uttarkand, Uttarpradesh and Madhaya Pradesh have legally permitted to cultivate Marijuana • The active principle in marijuana is transtetra hydrocanabinal (THC). • It is an effective pain reliever and reduces hypertension. • THC is used in treating Glaucoma a condition in which pressure develops in the eyes. • THC is also used in reducing nausea of cancer patients undergoing radiation and chemotherapy. • THC provides relief to bronchial disorders, especially asthma as it dilates bronchial vessels.	**(1)** **(1)** **(1)**	

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Part – II (Bio-Zoology) 35 marks

I. Choose the correct answer: 8 x 1 = 8

1. Down's syndrome is a genetic disorder which is caused by the presence of an extra chromosome number.
   a) 20 b) 21 c) 4 d) 23
2. Assertion (A) : Ovulation is the release of ovum from the Graffian follicle
   Reason (R) : It occurs during the follicular phase of the menstrual cycle.
   a) A and R are true, R is the correct explanation of A
   b) A and R are true, R is not the correct explanation of A
   c) A is true R is false
   d) Both A and R are false
3. An operon is a
   a) Protein that suppresses gene expression
   b) Protein that accelerates gene expression
   c) Cluster of structural genes with related function
   d) Gene that switched other genes on or off
4. Match the pathogens with respective diseases caused by them and select the correct match using the codes given below.

A.	Leishmania donavani	– i)	Amoebiasis
B.	Wuchereria bancrofti	– ii)	Kala–azar
C.	Trypanosoma gambiense	– iii)	Sleeping sickness
B.	Entamoeba histolytica	– iv)	Filariasis

   a) A–ii, B–iv, C–iii D–i b) A–ii, B– iv c–i, D–iii
   c) A–iii B–i, c–ii, D–iv d) A–i B–iv, C–iii. D–ii
5. The phenomenon of Industrial Melanism demonstrates a) Natural selection
   b) Induced mutation c) Reproductive isolation d) Geographical isolation
6. Predation and Parasitism are which type of interactions
   a) (+,+) b) (+,0) c) (–,–) d) (+,–)
7. The organisation which published the red list of species is
   a) WWF b) IUCN c) ZSI d) UNEP
8. Average human consumption of Oxygen per day
   a) 1000 L b) 1500 L c) 550 L d) 500 L

II. Answer any four questions. 4 x 2 = 8

9. What is plasmotomy?
10. What is chorionic villus sampling?
11. Why are sex linked recessive characters are more common in the male human beings?
12. Which is called as Brewer's yeast? What are its uses?
13. What are transgenic animals? Give examples
14. Name, the active chemical found in the medicinal plant Rauwolfia Vomitoria. What type of diversity it belongs to?

III. Answer any three questions. (Q.No.19 is compulsory) 3 x 3 = 9

15. Why tRNA is called an adapter molecule?
16. How does Neanderthal man differ from the modern man in appearence?
17. Draw a neat labelled diagram of HIV–virus.
18. Write any three functions of soil.
19. What is plastic waste? What are the remedies for plastic waste.

IV. Answer all the questions. 2 x 5 = 10

20. a) List out the major reproductive events in human beings. *(OR)*
    b) Why do we find a decrease in biodiversity distribution, if we move from the tropics towards the poles.
21. a) List out the salient features of genetic code. *(OR)*
    b) What are the various measures to control Air Pollution.

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TENTATIVE ANSWER KEY

CHOOSE THE CORRECT ANSWER (8 × 1 = 8)

Q.NO	ANSWER	MARKS
1	(b) 21	1
2	(c) A true R is false	1
3	(c) Cluster of structural genes with related function	1
4	(a) A-ii B –iv C-iii D-i	1
5	(a) Natural Selection	1
6	(d) (+,-)	1
7	(b) IUCN	1
8	(c) 550L	1

PART – II **(4 × 2 = 8)**

Answer any FOUR from the following question.

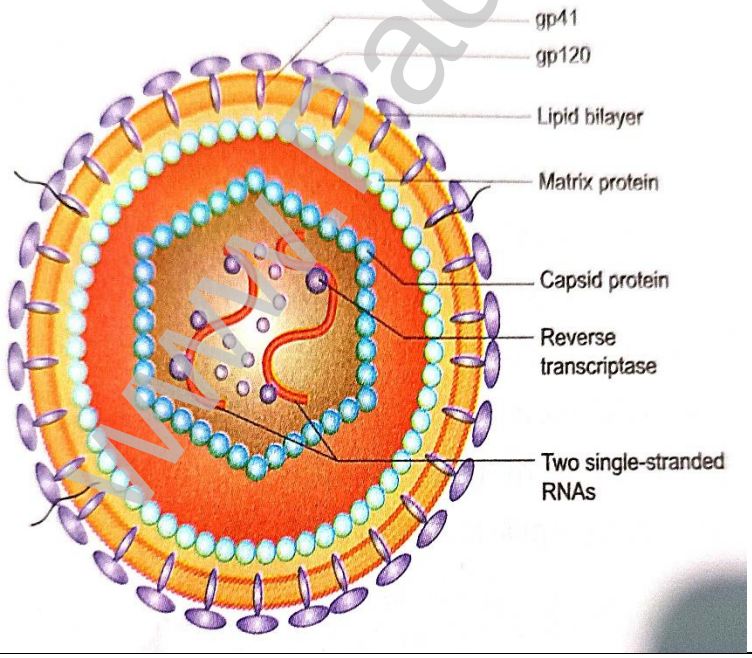
Q.NO	ANSWER	MARKS SPLITT	TOTAL MARKS
9	**What is plasmotomy ?** Plasmotomy is the division of multinucleated parent into many multinucleate daughter individuals with the division of nuclei	(2)	(2)
10	**Chorionic Villus Sampling (CVS)** CVS is a prenatal test that involves taking a sample of the placental tissue to test for chromosomal abnormalities	(2)	(2)
11	**Why are sex linked recessive characters more common in the male human beings?** Sex linked inherited traits are more common in males than females because, males are **hemizygous** and therefore express the trait when they inherit one mutant allele.	(2)	(2)
12	**Which is called as Brewer yeast? What are its uses?** i) ***Saccharomyces cerevisiae*** commonly called brewer's yeast. ii) It is used for fermenting malted cereals and fruit juices to produce various alcoholic beverages	(2)	(2)
13	**What are transgenic animals? Give examples.** The foreign DNA that is introduced is called the transgene and the animals that are produced by DNA manipulations are called transgenic animals or the genetically engineered or genetically modified organisms **E.g** ; mice, rat, rabbit, pig, cow, goat, sheep and fish	(2)	(2)
14	**Name the active chemical found in the medicinal plant Rauwolfia vomitoria. What type of diversity it belongs to?** i) Rouwolfia vomitaria, a medicinal plant growing in different ranges of the Himalayas	(2)	(2)

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	ii) It shows differences in the potency and concentration of the active ingredient **reserpine** due to genetic diversity.		

PART – III **(3 × 3 = 9)**

Answer any THREE from the following question. (Q.NO.19 is compulsory Question)

Q.NO	ANSWER	MARKS SPLITT	TOTAL MARKS
15	**Why tRNA is called an adapter molecule?** i) The transfer RNA, (tRNA) molecule of a cell acts as a vehicle that picks up the amino acids scattered through the cytoplasm and also reads specific codes of mRNA molecules. Hence it is called an adapter molecule. ii) This term was postulated by Francis Crick.	(3)	(3)
16	**How does Neanderthal man differ from the modern man in appearance** i) Neanderthal human was found in Neander Valley, Germany with a brain size of 1400 cc and lived between 34,000 - 1,00,000 years ago. ii) They differ from the modern human in having semierect posture, flat cranium, sloping forehead, thin large orbits, heavy brow ridges, protruding jaws and no chin. iii) They used animal hides to protect their bodies, knew the use of fire and buried their dead. They did not practice agriculture and animal domestication.	(1+1+1)	(3)
17	Structure of HIV  	(1+2=3))	(3)
18	**Write any three functions of soil ?** • medium for plant growth • means for water storage and purification • modifier of earth's atmosphere	(1+1+1=3	(3)

19 (C.Q)	**What is plastic waste? What are the remedies of plastic waste?** i) Plastics are low molecular weight organic polymers that are non-degradable in the natural environment. ii) They are used in several items, including cars, bulletproof vests, toys, hospital equipment, carry bags and food containers. **Remedies :** • Remedies: '4R'- Refuse, Reduce, Reuse and Recycle mantra is the best available remedy for plastic waste pollution.	(1+2=3)	(3)

PART – IV (2 × 5 = 10)

Answer ALL from the following question

Q.NO	ANSWER	MARKS SPLITT	TOTAL MARKS
20 (a)	**Explain the major reproductive events in human beings ?** i) **Gametogenesis:** Formation of gametes by spermatogenesis and oogenesis. ii) **Insemination**: Transfer of sperms by the male into the female genital tract. iii) **Fertilization**: Fusion of male and female gametes to form zygote, called Fertilization. iv) **Cleavage**: Rapid mitotic divisions of the zygote which convert the single celled zygote into a multicellular structure called blastocyst. v) **Implantation**: Attachment of blastocyst to the uterine wall. vi) **Placentation:** Formation of placenta which is the intimate connection between foetus and uterine wall of the mother for exchange of nutrients. vii) **Gastrulation**: Process by which blastocyst is changed into a gastrula with three primary germ layers. viii) **Organogenesis:** Formation of specific tissues, organs and organ systems from three germ layers. ix) **Parturition**: Expulsion of the foetus from the mother's womb	(1) (1) (1) (1) (1)	**(5)**
20 (b)	**Why do we find a decrease in biodiversity distribution, if we move from the tropics towards the poles?** i) The most important pattern of biodiversity is latitudinal gradient in diversity. ii) This means that there is an increasing diversity from the poles to equator. iii) Diversity increases as one moves towards the temperate zones and reaches the maximum at the tropics. iv) Thus, tropics harbour more biodiversity than temperate or polar regions, especially between the latitudes of 23.5^{o}N and 23.5^{o}S (Tropic of Cancer to the Tropic of Capricorn). v) Harsh conditions exist in temperate areas during the cold seasons while very harsh conditions prevail for most of the year in polar regions.	**(1)** **(1)** **(1)** **(1)** **(1)**	**(5)**
21 (a)	**Write the salient features of genetic code ?**		

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	i) The genetic codon is a triplet code and 61 codons code for amino acids and 3 codons do not code for any amino acid and function as stop codon (Termination). ii) The genetic code is universal. It means that all known living systems use nucleic acids and the same three base codons (triplet codon) direct the synthesis of protein from amino acids iii) A non-overlapping codon means that the same letter is not used for two different codons. iv) It is comma less, which means that the message would be read directly from one end to the other i.e., no punctuation are needed between two codes. v) The code is always read in fixed directions 5 →3 direction. vi) AUG has dual functions.It acts as a initiator codon and also codes for the amino acid methionine. vii) UAA, UAG, UGA are termination codon and also called Non sense codon.	**(1)** **(1)** **(1)** **(1)** **(1)**	**(5)**
21 (b)	**What are the various measures to control of Air pollution?** Certain measures help to remove pollutants, reduce their presence or prevent their entry into the atmosphere. • Trees are the best remedy for urban particulate and gaseous pollution • Forests act as carbon sinks and lungs of the planet • Catalytic converters in vehicles help to reduce polluting gases drastically • Diesel exhaust filters in automobiles cuts particulates • Electrostatic precipitators reduce release of industrial pollutants • Cost effective air pollution treatment systems like indoor plants and high performance biofilters can improve indoor air quality	**(1)** **(1)** **(1)** **(1)** **(1)**	**(5)**

ROUGH WORK